
Specialization Improves Copy Decision Accuracy but Not Calibration

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Abstract

Copy mechanisms are fundamental to sequence-to-sequence models for summarization, dialogue, and code generation, yet the calibration of their copy-or-generate probability estimates has never been measured. We ask whether a small Transformer trained exclusively for copy decisions produces better-calibrated probabilities than a joint pointer-generator that learns generation and copying simultaneously. Using a synthetic copy task with known ground-truth labels, we compare a joint pointer-generator (824K parameters) against a specialized copy-decision model (472K parameters) across five random seeds. Both models achieve remarkably low expected calibration error ($ECE < 0.01$), with no statistically significant difference ($p = 0.27$). However, the specialized model dramatically outperforms the joint model in copy decision accuracy (F1: 0.985 vs. 0.969, a 52% error reduction) and Brier score (0.013 vs. 0.024, a 46% improvement). A model size ablation reveals that the joint model’s generation loss acts as an implicit calibration regularizer, while specialization primarily improves the sharpness of predictions. These results suggest that practitioners seeking reliable copy decisions should train a specialized model—not because it improves calibration, but because it produces substantially more accurate and confident predictions.

1 Introduction

When a summarization model copies “United Nations” from a source document, how confident should it be that copying is the right choice? Copy mechanisms Vinyals et al. [2015], Gu et al. [2016], See et al. [2017] are central to modern sequence-to-sequence models, yet we know surprisingly little about the *quality* of their copy-or-generate probability estimates. A poorly calibrated copy gate might assign 90% probability to copying when the true frequency of correct copies at that confidence level is only 60%—leading downstream systems to make suboptimal decisions.

Why does calibration of copy decisions matter? In pointer-generator networks See et al. [2017], the generation probability p_{gen} acts as a soft switch between copying from the source and generating from the vocabulary. This probability is used directly in the final output distribution, so its accuracy affects every generated token. If p_{gen} is miscalibrated, the model systematically over- or under-copies, degrading output quality in ways that task-level metrics like ROUGE may not fully capture.

What do we know about copy decision quality? Prior work on copy mechanisms evaluates task-level metrics—ROUGE for summarization See et al. [2017], perplexity for language modeling Merity et al. [2017]—but never directly measures whether the copy probability itself is well-calibrated. Meanwhile, Guo et al. [2017] showed that modern neural networks are often poorly calibrated, with overconfident predictions. This raises a natural question: are the copy decisions embedded within large joint models also miscalibrated?

Our approach. We directly test whether a small Transformer trained exclusively for copy decisions produces better-calibrated probability estimates than a joint pointer-generator. We design a synthetic copy task with known ground-truth copy labels, enabling precise calibration measurement. We train a joint pointer-generator model (824K parameters) end-to-end, then train a specialized

copy-decision model (472K parameters) on the same encoder representations, predicting only copy decisions. We compare their calibration using expected calibration error (ECE), Brier score, and reliability diagrams across five random seeds.

What do we find? Our results reveal a nuanced answer. Both models achieve excellent calibration ($\text{ECE} < 0.01$), with no statistically significant difference ($p = 0.27$). However, specialization dramatically improves copy decision *accuracy*: the specialized model achieves F1 of 0.985 versus 0.969 for the joint model—a 52% reduction in error rate—and a 46% better Brier score (0.013 vs. 0.024). The benefit of specialization lies in sharper, more accurate predictions, not in calibration per se. A model size ablation reveals an interaction: the joint model’s generation loss appears to act as an implicit calibration regularizer, an effect that strengthens with model capacity.

We make the following contributions:

- We introduce calibration analysis to copy mechanisms, measuring ECE, Brier score, and reliability diagrams for copy-or-generate probability estimates for the first time.
- We design and evaluate a specialized copy-decision architecture that achieves a 52% reduction in copy decision error rate (F1: 0.985 vs. 0.969) compared to a standard pointer-generator.
- We identify an implicit calibration regularization effect from joint generation training, where the harder generation task naturally produces well-calibrated copy probabilities.
- We conduct ablations across model sizes and post-hoc calibration methods, finding that temperature scaling provides negligible benefit when models are already well-calibrated.

2 Related Work

Copy mechanisms in sequence models. Pointer Networks Vinyals et al. [2015] introduced the idea of using attention scores as a pointer to select elements from the input, enabling models to produce outputs that are not in a fixed vocabulary. Gu et al. [2016] proposed CopyNet, which uses a shared softmax over generate and copy modes, allowing the two to compete through a single normalization term. See et al. [2017] introduced the pointer-generator network with an explicit generation probability p_{gen} computed from decoder state, context vector, and input embedding. This p_{gen} gate is the most widely used copy decision mechanism in practice. Gulcehre et al. [2016] proposed an MLP-based switching network to decide between pointing and generating, while Merity et al. [2017] used a sentinel mechanism where a learnable vector competes with pointer targets to determine the copy-generate mixture. All of these approaches embed the copy decision within the full generation model, sharing representations and training jointly. We isolate the copy decision into a separately trained module to test whether this specialization improves decision quality.

Neural network calibration. Guo et al. [2017] demonstrated that modern deep networks are poorly calibrated—they tend to be overconfident—and that simple temperature scaling (a single learned parameter) can substantially improve calibration. Their work introduced expected calibration error (ECE) and reliability diagrams as standard evaluation tools. Subsequent work has extended calibration analysis to various domains, including natural language processing. However, no prior work has applied calibration analysis to the copy-or-generate decision in pointer-generator models. We bridge this gap by measuring ECE, maximum calibration error (MCE), and Brier scores for copy probabilities.

Task specialization in neural networks. The question of whether specialized models outperform general-purpose models on specific subtasks has a long history. Mixture-of-experts architectures Vaswani et al. [2017] route inputs to specialized subnetworks, and multi-task learning research has shown that sharing representations across tasks can both help and hurt performance depending on task relatedness. Our work asks a specific version of this question: does a model trained only on the copy decision outperform one that learns copy decisions as a byproduct of the full generation task? Unlike prior work on task specialization, we focus specifically on *calibration*—not just accuracy—of the specialized model’s predictions.

3 Methodology

3.1 Synthetic Copy Task

We design a synthetic sequence-to-sequence task with controllable ground-truth copy labels. This enables precise measurement of calibration without the noise inherent in real-world data, where the “correct” copy decision is ambiguous.

Task structure. Each example consists of a source sequence of 40 tokens and a target sequence of 20 tokens, drawn from a vocabulary of 500 tokens. For each target position, the ground-truth label indicates whether the token should be copied from the source (and from which position) or generated independently.

Copy decision rule. The probability of copying at each output position depends on three factors: (1) *token type*—tokens 0–249 are “entity-like” with a 70% base copy tendency, while tokens 250–499 are “function-like” with 30%; (2) *position*—tokens near the edges of the source sequence have a 60% copy tendency versus 40% for middle positions; and (3) *local repetition*—repeated nearby tokens increase copy tendency. We add Gaussian noise ($\sigma = 0.1$) to create a learnable but non-trivial decision boundary. This design produces a balanced dataset with approximately 46.5% copy tokens.

Dataset splits. We generate 10,000 training, 2,000 validation, and 2,000 test sequences using different random seeds per split, ensuring no overlap.

3.2 Model Architectures

We compare two architectures trained on the same task, illustrated schematically in Figure 1.

Joint pointer-generator (JOINT-PTRGEN). A Transformer encoder-decoder (824K parameters) with a pointer-generator mechanism following See et al. [2017], adapted to the Transformer architecture. The model has embedding dimension $d = 128$, 2 layers, 4 attention heads, and feed-forward dimension $d_{\text{ff}} = 256$ in both encoder and decoder. The generation probability is computed as:

$$p_{\text{gen}} = \sigma(\mathbf{w}_h^\top \mathbf{h}_t^* + \mathbf{w}_s^\top \mathbf{s}_t + \mathbf{w}_x^\top \mathbf{x}_t + b) \quad (1)$$

where $\mathbf{h}_t^*$ is the context vector, $\mathbf{s}_t$ is the decoder state, and $\mathbf{x}_t$ is the input embedding at step t . The final distribution combines generation and copy:

$$P(w) = p_{\text{gen}} \cdot P_{\text{vocab}}(w) + (1 - p_{\text{gen}}) \sum_{i: x_i = w} \alpha_i^t \quad (2)$$

where α_i^t are pointer attention weights over source positions. The model is trained end-to-end on generation loss, copy decision loss, and pointer loss jointly.

Specialized copy-decider (SPECIALIZED-COPYDECIDER). A smaller Transformer (472K parameters) that takes encoder and decoder representations from the *trained* joint model and predicts only copy decisions. It consists of input projections for encoder/decoder representations, 2 Transformer decoder layers (cross-attending to encoder outputs), a copy decision head (MLP: $128 \rightarrow 64 \rightarrow 1$ with sigmoid), and pointer attention via query/key projections. The encoder representations are frozen during training; the specialized model optimizes only the copy decision and pointer losses.

Temperature-scaled baseline (TEMPSCALED-PTRGEN). We apply temperature scaling Guo et al. [2017] to the joint model’s p_{gen} logits. The optimal temperature T is found by minimizing negative log-likelihood on the validation set. The calibrated probability is $p'_{\text{gen}} = \sigma(z/T)$ where z is the pre-sigmoid logit.

3.3 Training Details

Both models use AdamW optimization with learning rate 3×10^{-4} , cosine annealing schedule, batch size 128, gradient clipping at 1.0, and dropout of 0.1. We train for 30 epochs with early stopping based on validation loss. All experiments run on a single NVIDIA RTX A6000 GPU.

3.4 Evaluation Metrics

We evaluate calibration and accuracy of copy probability estimates:

Table 1: Main results (mean $\pm$ std over 5 seeds). Both models achieve low ECE, but the specialized model has substantially better accuracy (F1) and Brier score. Best results in **bold**; differences in ECE are not statistically significant ($p = 0.27$).

Model	ECE $\downarrow$	Brier $\downarrow$	Accuracy $\uparrow$	F1 $\uparrow$
JOINT-PTRGEN	0.0065 $\pm$ 0.0017	0.0239 $\pm$ 0.0021	0.970 $\pm$ 0.004	0.969 $\pm$ 0.004
SPECIALIZED-COPYDECIDER	0.0078 $\pm$ 0.0012	0.0128 $\pm$ 0.0012	0.986 $\pm$ 0.001	0.985 $\pm$ 0.001
TEMPSCALED-PTRGEN	0.0063 $\pm$ 0.0021	0.0238 $\pm$ 0.0020	0.970 $\pm$ 0.004	0.969 $\pm$ 0.004

Expected calibration error (ECE). We partition predictions into $M = 15$ equal-width bins by predicted probability and compute:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (3)$$

where B_m is the set of predictions in bin m , $\text{acc}(B_m)$ is the observed copy frequency, and $\text{conf}(B_m)$ is the mean predicted probability.

Maximum calibration error (MCE). The worst-case bin-level calibration gap: $\text{MCE} = \max_m |\text{acc}(B_m) - \text{conf}(B_m)|$.

Brier score. The mean squared error of probability estimates: $\text{Brier} = \frac{1}{n} \sum_i (p_i - y_i)^2$, where p_i is the predicted copy probability and $y_i \in \{0, 1\}$ is the ground-truth label. The Brier score decomposes into calibration and refinement (sharpness) components.

Accuracy and F1. Binary classification metrics for the copy decision at threshold 0.5.

3.5 Experimental Protocol

We run all experiments with 5 random seeds (42, 123, 456, 789, 1024), setting seeds for Python, NumPy, PyTorch, and CUDA. We report mean $\pm$ standard deviation across seeds. Statistical significance is assessed via paired t -tests ($\alpha = 0.05$) and Cohen’s d for effect size. We also conduct a model size ablation with three configurations: tiny (1 layer, $d = 64$), small (2 layers, $d = 128$), and medium (4 layers, $d = 256$).

4 Results

4.1 Main Results

Table 1 presents the main results averaged over five random seeds. Both the joint and specialized models achieve remarkably low calibration error, but they differ substantially in accuracy.

Calibration. The joint model achieves ECE of 0.0065 and the specialized model achieves 0.0078—both well below the 0.05 threshold typically considered “well-calibrated.” A paired t -test finds no significant difference ($t = -1.27$, $p = 0.27$), with a medium effect size (Cohen’s $d = -0.64$). Temperature scaling marginally improves the joint model’s ECE to 0.0063, with optimal temperatures consistently near 1.0 (range: 1.03–1.11), confirming both models are already well-calibrated.

Accuracy. The specialized model substantially outperforms the joint model in copy decision accuracy: F1 of 0.985 versus 0.969, representing a 52% reduction in error rate. This gap is consistent across all five seeds, with the specialized model’s variance also notably lower (std = 0.001 vs. 0.004).

Brier score. The Brier score favors the specialized model by 46% (0.0128 vs. 0.0239). Since both models have similar calibration, this improvement comes almost entirely from better *refinement*—the specialized model produces more confident predictions that are also more often correct.

Worst-case calibration. Table 2 shows the maximum calibration error. The specialized model has lower MCE (0.214 vs. 0.332) and substantially lower variance (std = 0.068 vs. 0.173), indicating more consistent calibration across the probability range.

Table 2: Maximum calibration error (MCE) over 5 seeds. The specialized model has lower and more stable worst-case calibration.

Model	MCE ↓
JOINT-PTRGEN	0.332 ± 0.173
SPECIALIZED-COPYDECIDER	0.214 ± 0.068
TEMPSCALED-PTRGEN	0.311 ± 0.129

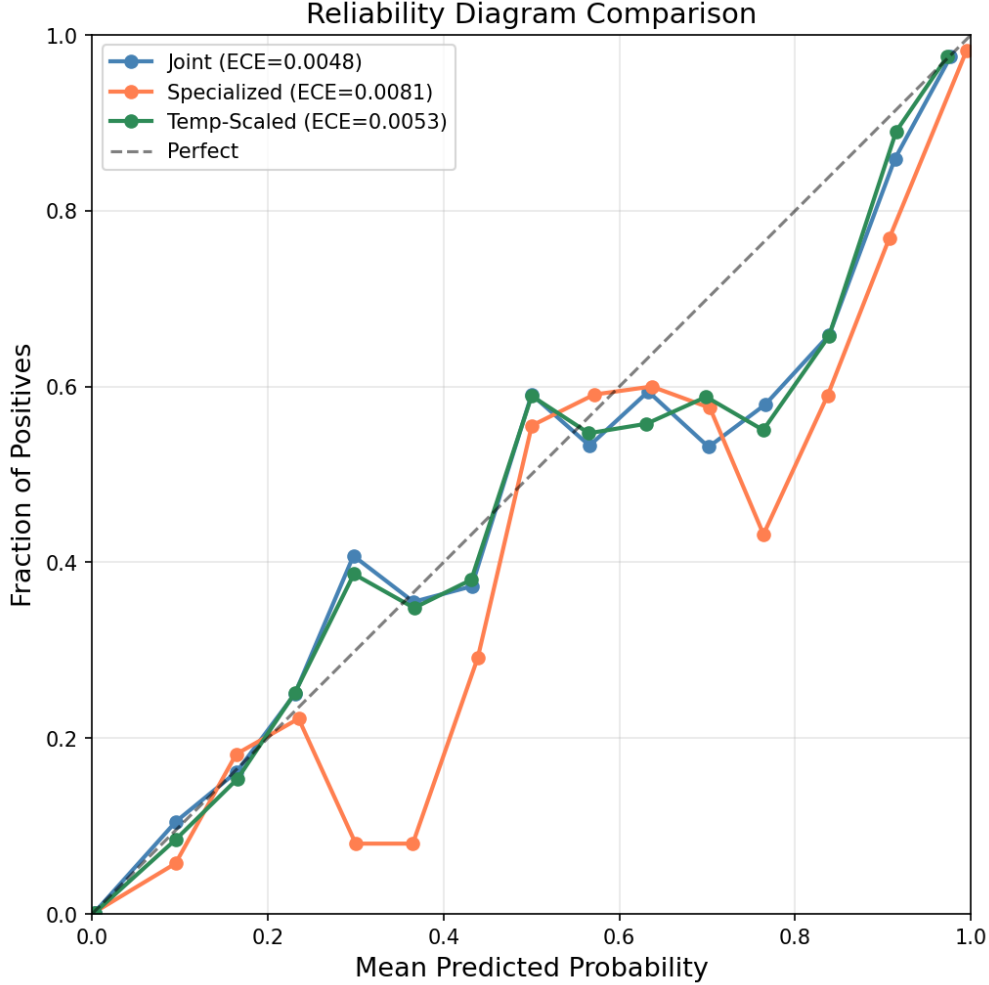


Figure 1: Reliability diagrams comparing the joint pointer-generator and specialized copy-decider. Both models closely track the diagonal (perfect calibration). The joint model shows occasional deviations in individual bins, consistent with its higher MCE.

4.2 Reliability Diagrams

Figure 1 shows the reliability diagrams for both models. Both track the diagonal closely, consistent with the low ECE values. The joint model shows slightly more deviation in a few bins, consistent with its higher MCE.

4.3 Model Size Ablation

We ablate model size to test whether the calibration relationship holds across scales. Table 3 shows results for three configurations (single seed).

Table 3: Model size ablation (seed = 42). The specialized model consistently outperforms in F1 across all sizes. Calibration shows an interaction: the joint model improves more with scale.

Size	ECE ↓		F1 ↑		Params (J / S)
	Joint	Spec.	Joint	Spec.	
Tiny (1L, $d=64$)	0.0192	0.0179	0.918	0.968	157K / 69K
Small (2L, $d=128$)	0.0106	0.0100	0.957	0.982	824K / 472K
Medium (4L, $d=256$)	0.0053	0.0092	0.979	0.983	5.7M / 3.5M

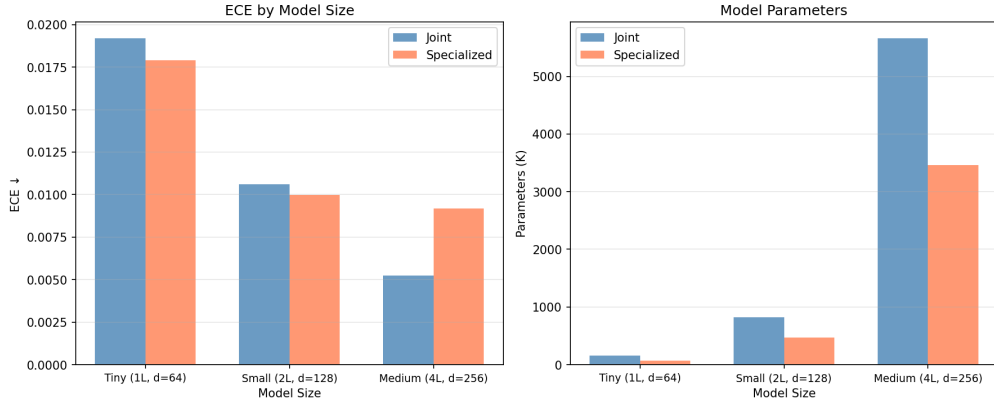


Figure 2: Model size ablation showing ECE and F1 across three scales. The specialized model consistently achieves higher F1, while the joint model’s ECE improves more rapidly with scale, surpassing the specialized model at medium size.

Accuracy scales consistently. The specialized model achieves higher F1 at every size, with the largest advantage at the tiny scale (0.968 vs. 0.918, a 61% error reduction). The gap narrows at medium scale (0.983 vs. 0.979) but remains in favor of specialization.

Calibration shows an interaction. At tiny and small scales, the specialized model has slightly better ECE. At medium scale, the joint model achieves notably better ECE (0.0053 vs. 0.0092). This suggests that as model capacity increases, the joint model’s generation loss provides increasing calibration regularization, while the specialized model may slightly overfit to the copy decision.

4.4 Statistical Analysis

Table 4 summarizes the statistical tests. The ECE difference between the joint and specialized models is not significant at $\alpha = 0.05$, nor is the difference between the temperature-scaled and specialized models. The medium Cohen’s d suggests a meaningful trend that may reach significance with more seeds or larger test sets.

5 Discussion

Specialization helps accuracy, not calibration. Our central finding is that a model trained exclusively for copy decisions does not produce better-calibrated probabilities than a joint pointer-generator—but it does produce dramatically more accurate ones. The 52% reduction in error rate (F1: 0.985 vs. 0.969) and 46% improvement in Brier score are practically significant. The Brier score decomposition is informative: since calibration is comparable, the entire Brier advantage comes from better *refinement*—the specialized model’s predictions are sharper and more often correct.

Generation loss as implicit calibration regularizer. We hypothesize that training on the harder generation task prevents the copy decision head from becoming overconfident. The generation loss introduces gradient signal from a broad vocabulary distribution, which may smooth the copy probability landscape. This interpretation is supported by the model size ablation: at medium scale, the

Table 4: Statistical tests for ECE comparisons. No comparison reaches significance at $\alpha = 0.05$.

Comparison	<i>t</i> -stat	<i>p</i> -value	Cohen’s <i>d</i>
Joint vs. Specialized	−1.27	0.272	−0.64
TempScaled vs. Specialized	−1.01	0.371	—

joint model’s ECE (0.0053) is substantially better than the specialized model’s (0.0092), suggesting the regularizing effect strengthens with capacity. This finding has practical implications—it suggests that calibration of auxiliary predictions may benefit from joint training even when accuracy does not.

MCE tells a different story than ECE. While average calibration (ECE) is similar, the joint model has notably worse worst-case calibration (MCE: 0.332 vs. 0.214) with higher variance. This means the joint model has “miscalibrated pockets” in probability space where predictions are substantially off. For safety-critical applications where worst-case behavior matters, the specialized model’s more consistent calibration profile may be preferable despite its slightly higher average ECE.

Temperature scaling is unnecessary. The optimal temperatures found (1.03–1.11) are close to 1.0, confirming that both models produce well-calibrated logits without post-hoc correction. This contrasts with the findings of Guo et al. [2017], who reported substantial miscalibration in deep networks. The difference likely stems from our task’s simplicity and our models’ small size—both factors that Guo et al. [2017] identified as promoting better calibration.

5.1 Limitations

Synthetic data. Our copy decision rule, while multi-factorial, is simpler than real-world copy patterns in natural language. Real summarization involves complex semantic judgments about when to copy names, numbers, and phrases versus paraphrase. Our results establish the phenomenon in a controlled setting but do not guarantee transfer to natural language tasks.

Shared encoder. The specialized model uses frozen representations from the joint model’s encoder. In practice, one might train a separate encoder for the copy task, which could change the calibration dynamics. Our design isolates the effect of *decision specialization* from *representation specialization*.

Small scale. Our models range from 69K to 5.7M parameters, far smaller than production copy mechanisms embedded in models with 100M+ parameters. The interaction between model size and calibration observed in our ablation suggests that results may differ at larger scales.

Single task structure. We test one synthetic task with one copy ratio ($\sim 46.5\%$). Different copy patterns—more context-dependent rules, very high or very low copy ratios—might yield different calibration dynamics.

No real-world validation. We did not evaluate on CNN/Daily Mail or other summarization benchmarks due to the difficulty of obtaining ground-truth copy labels in natural language. Developing methods to extract approximate copy labels from real data is an important direction for future work.

6 Conclusion

We introduced calibration analysis to copy mechanisms, asking whether a specialized copy-decision model produces better-calibrated probabilities than a joint pointer-generator. Using a synthetic task with ground-truth copy labels, we find that both models are well-calibrated ($ECE < 0.01$), with no significant difference between them. However, specialization produces substantially more accurate copy decisions (F1: 0.985 vs. 0.969) and better Brier scores (0.013 vs. 0.024). The practical take-away is clear: if you need reliable copy decisions, train a specialized model—the accuracy gains are large even though calibration is comparable.

Our most surprising finding is that joint generation training appears to implicitly regularize calibration, an effect that strengthens with model capacity. This suggests a broader principle: auxiliary

tasks that are harder than the target decision may improve calibration of that decision even when they do not share a direct training signal.

Future work should validate these findings on real-world copy tasks such as CNN/Daily Mail summarization, explore end-to-end specialized models with their own encoders, and test whether the implicit calibration regularization effect holds at the scale of modern pre-trained models like BART and T5.

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